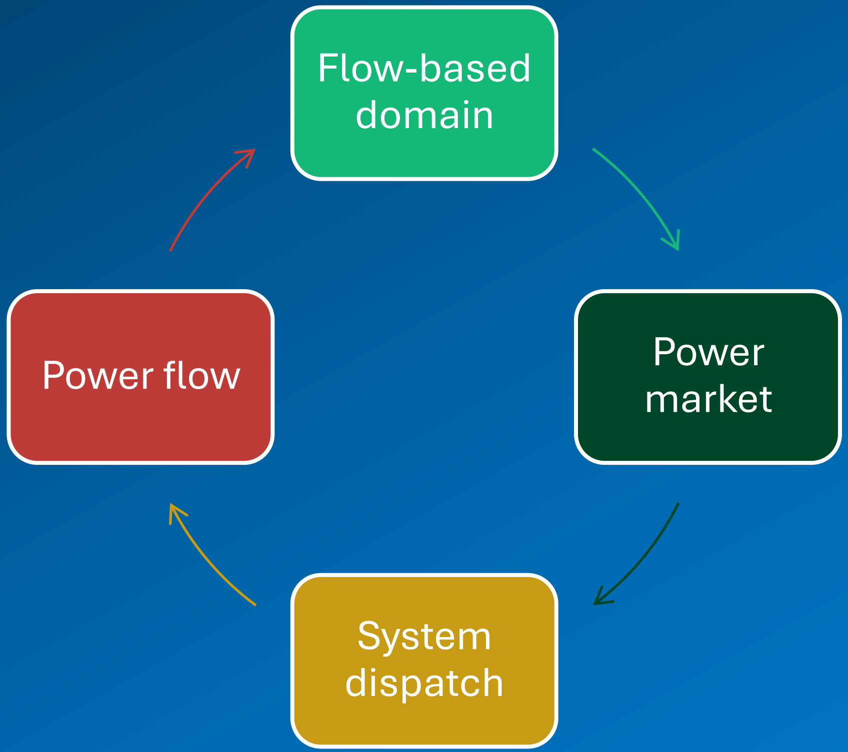


Here and back again: From details to aggregation

and the mystery of Generation Shift Keys

Stefan Jaehnert, Sintef Energi

Outline

CALCULATION OF THE
FLOW-BASED DOMAIN

FROM NODAL TO ZONAL
REPRESENTATION

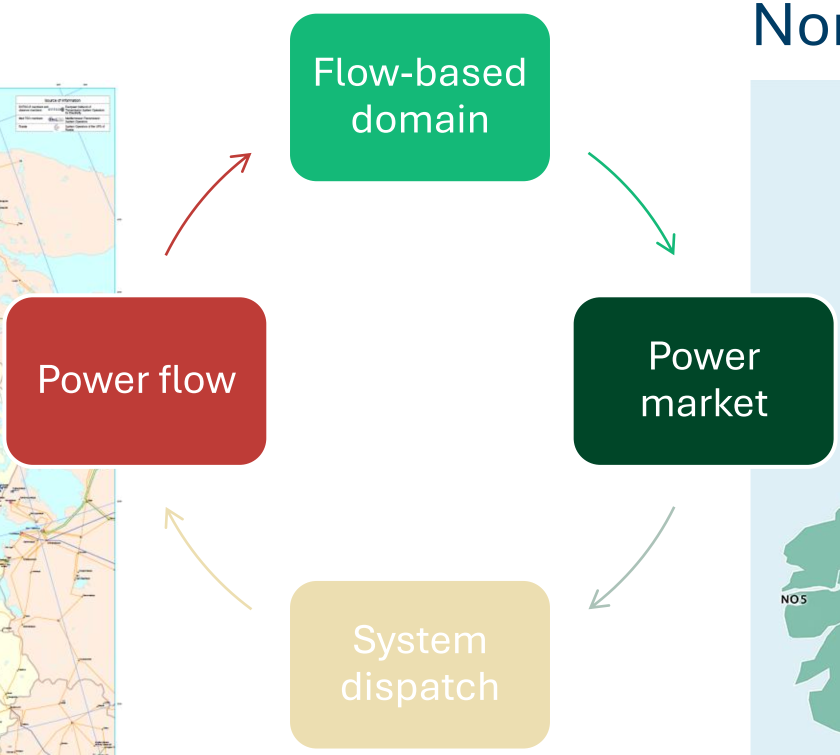
GENERATION SHIFT
KEYS

SELECTED ANALYSES
RESULTS

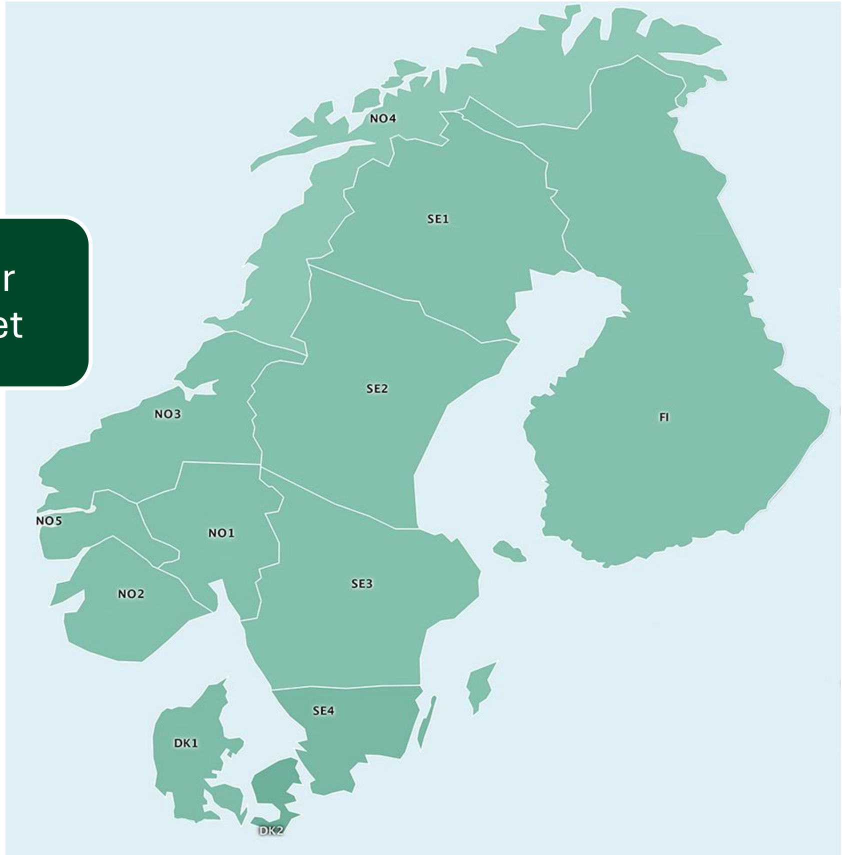
Interaction between the grid and price areas

or from DC power flow to PTDF domain

Nordic grid



Nordic price areas



Flow based market coupling - schedule

D-2
Base case forecast

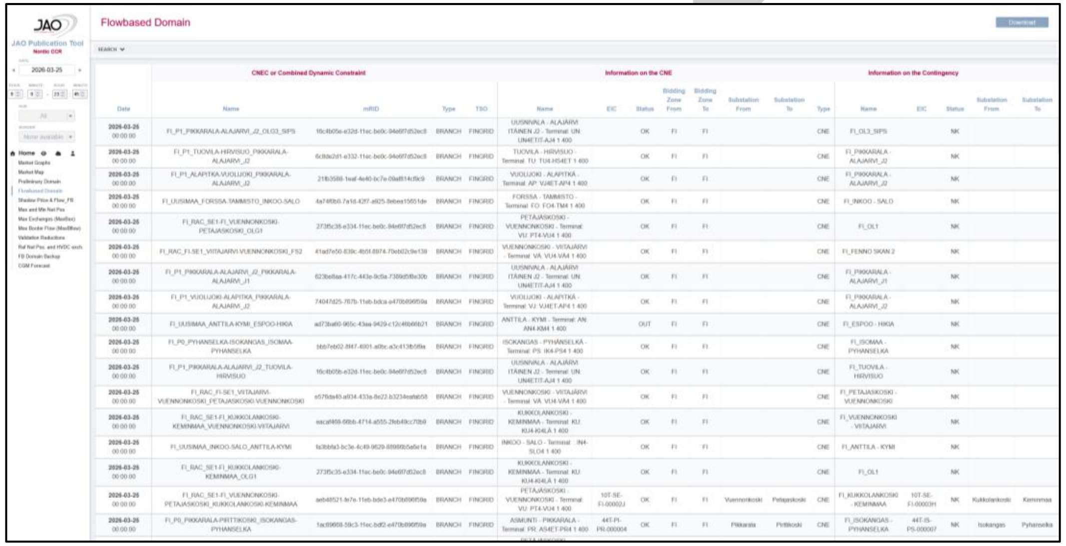
- Dispatch forecast
- Reference flows
- Reference net positions
- FBMC domain

D-1
FB market coupling

- Market solution and dispatch
- Market exchange

D-0
Real-time operation

- Physical flow
- Redispatch and balancing



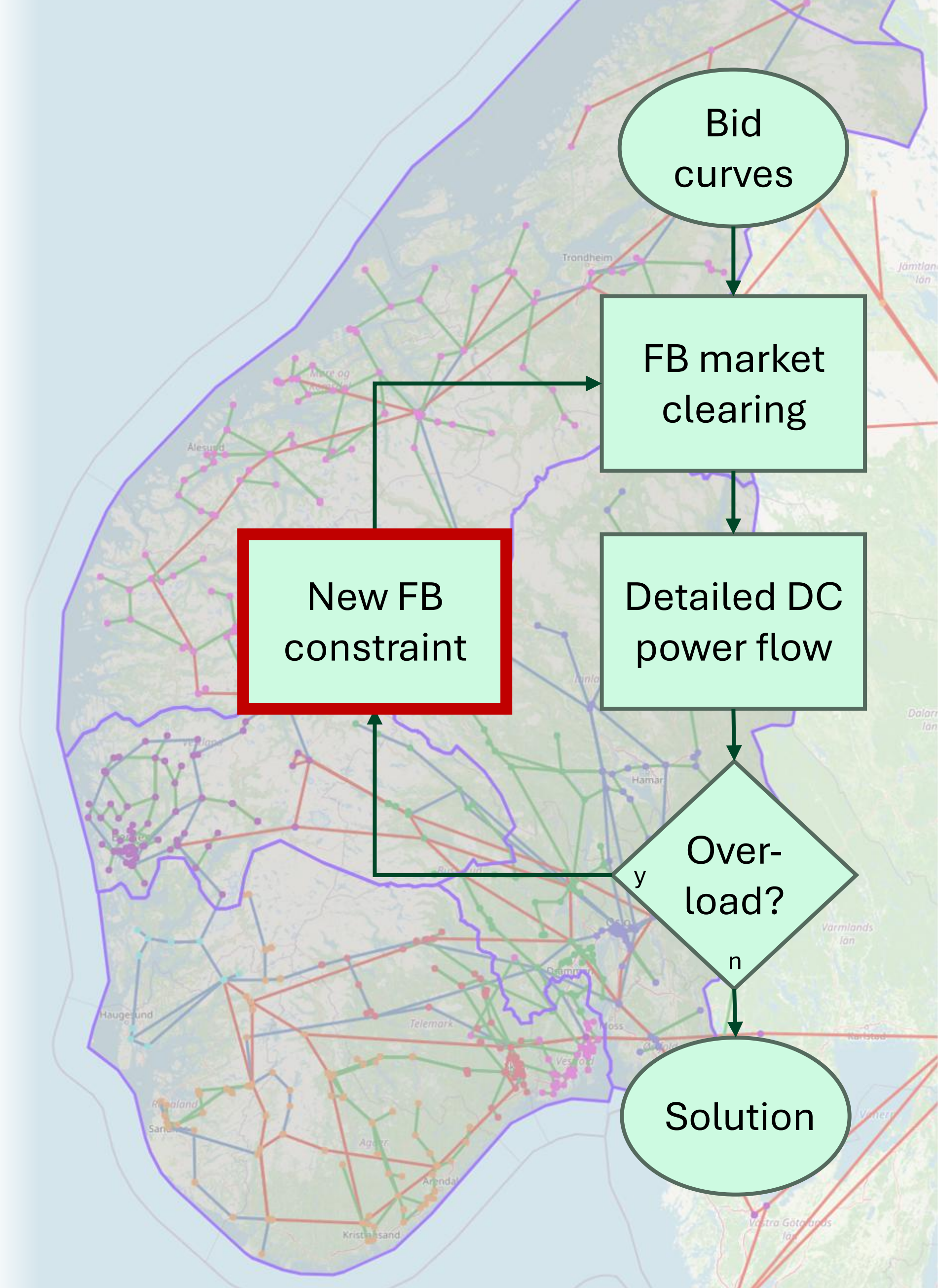
https://publicationtool.jao.eu/nordic/flowbasedDomain

Samnett modelling tool

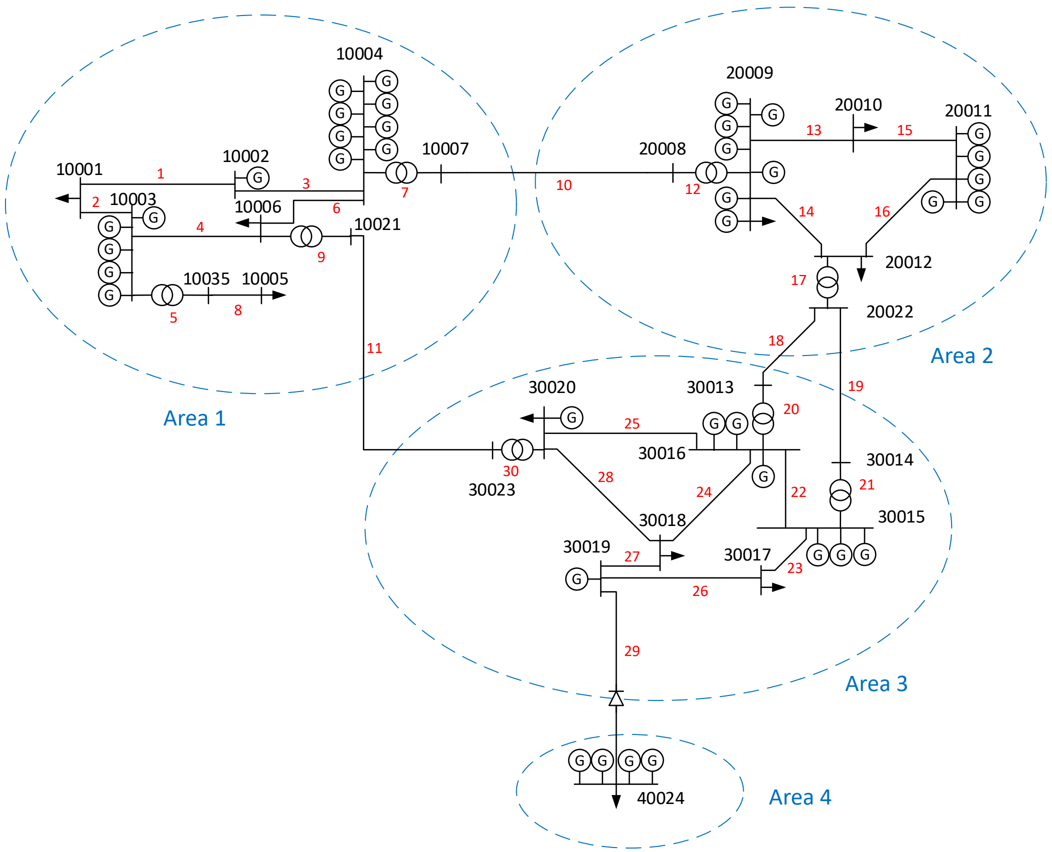
simulating flow based markets

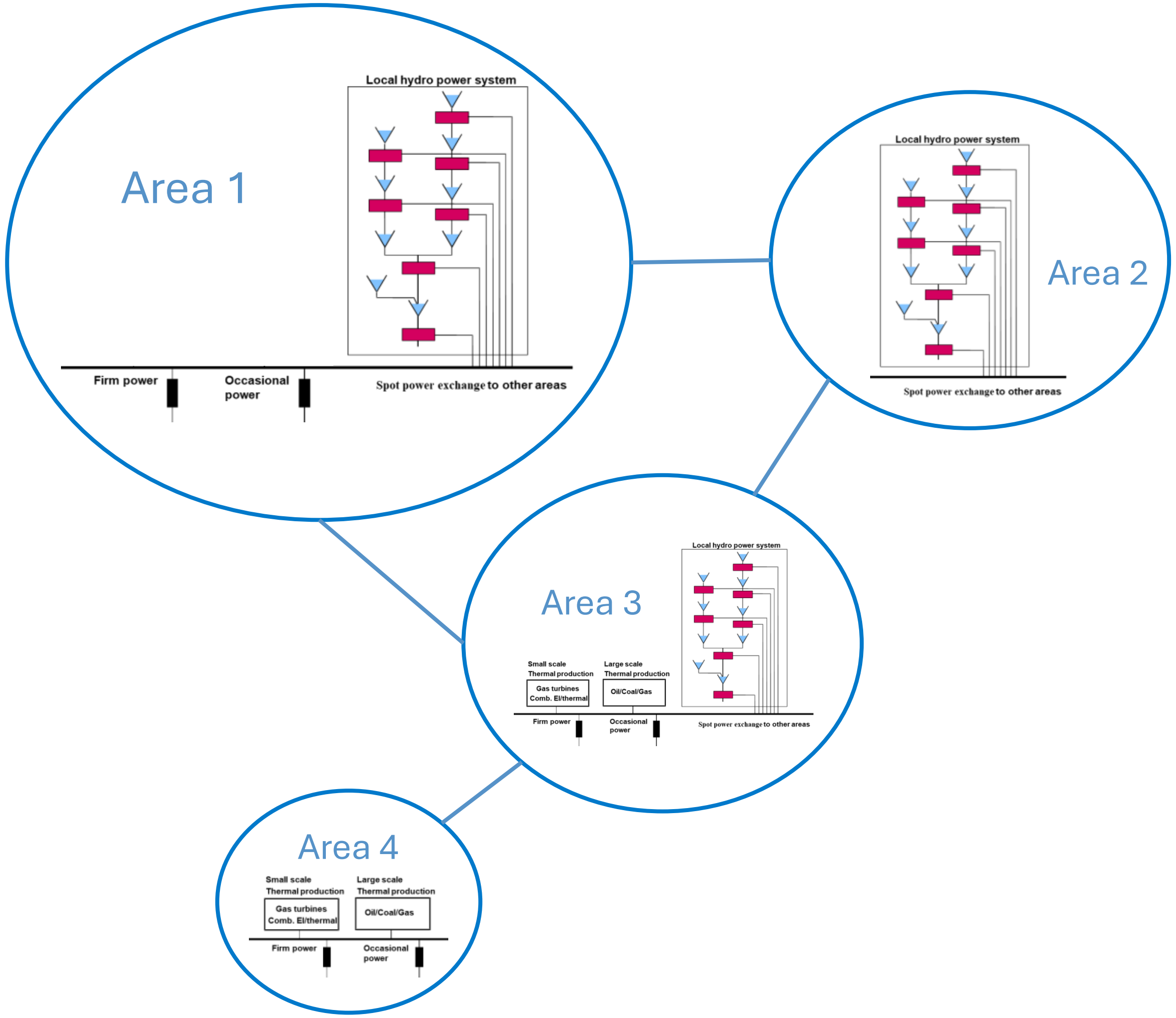
- Power market simulator accounting for detailed flow, by calculating and applying endogenous flow-based constraints
- Developed during 2010-2012 as part of a NFR research project and used by Statnett, Svenska Kraftnät and NVE
- Iterations between market clearing and power flow calculation to estimate all relevant flow-based constraints

Samnett - SINTEF's power market model with detailed grid



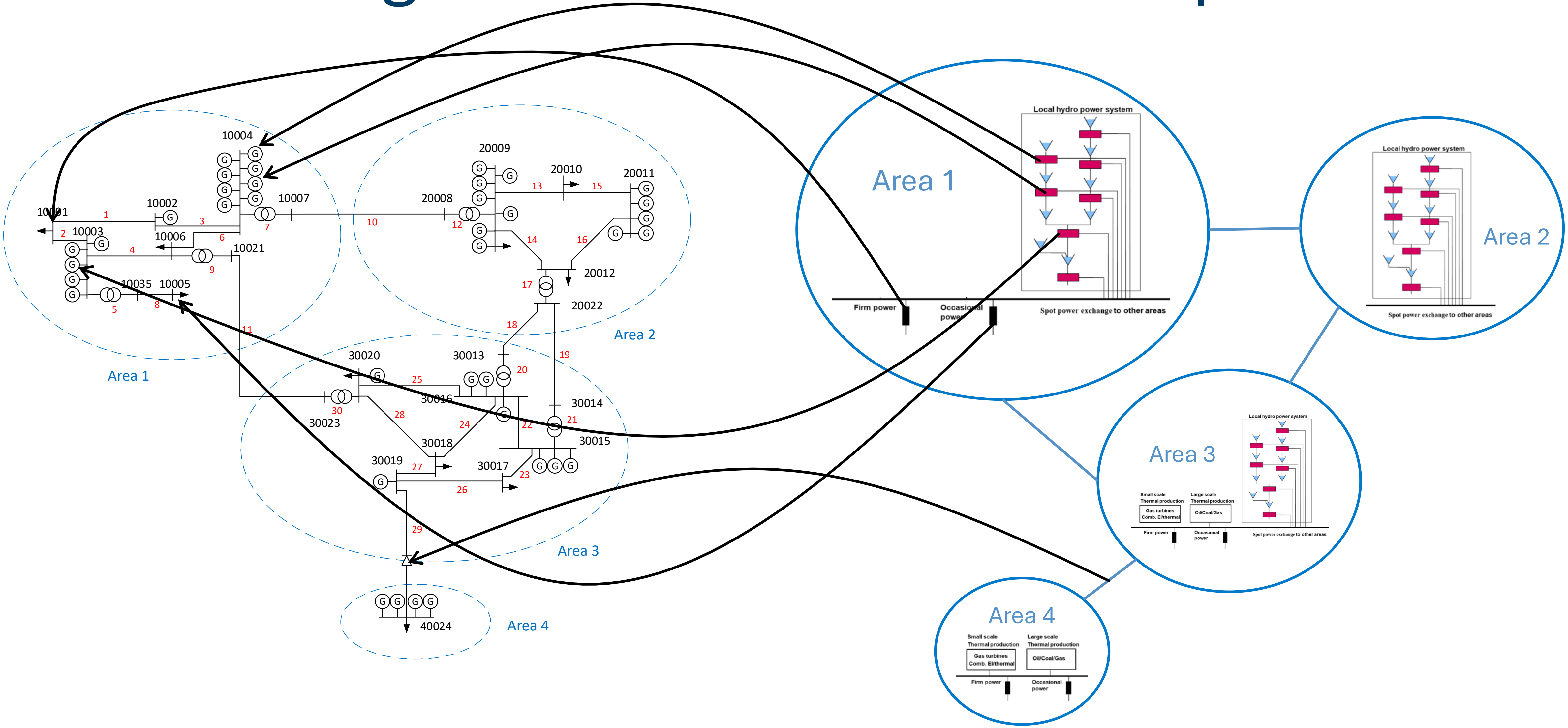
Market and grid simulation



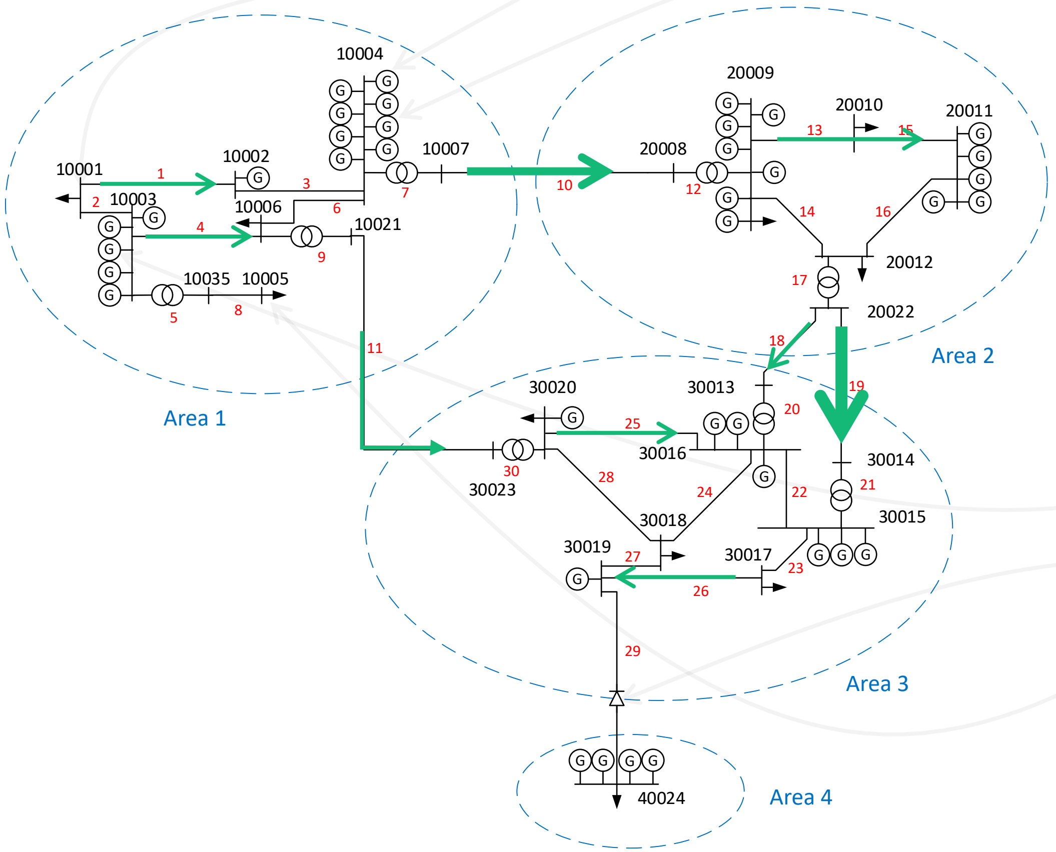


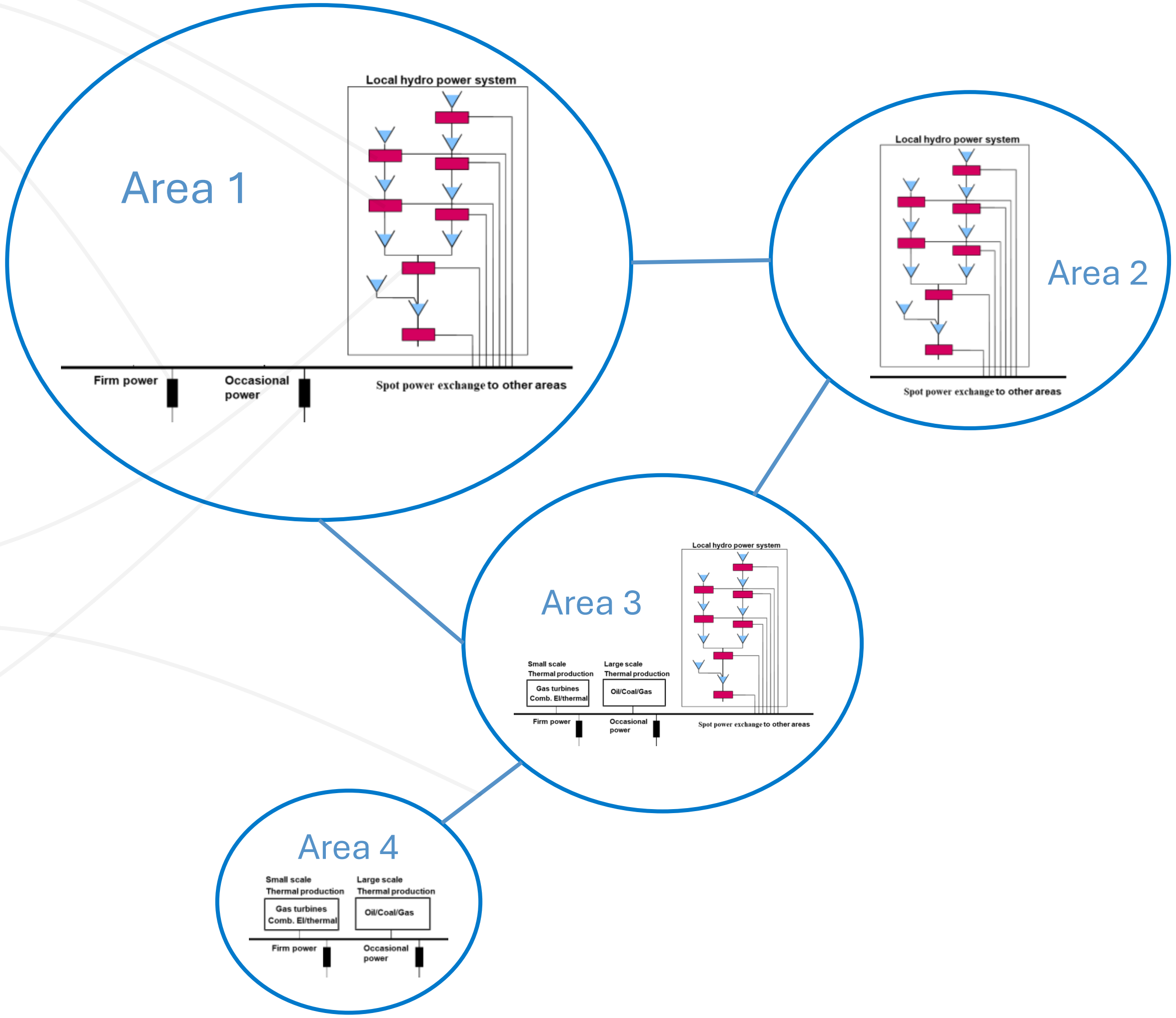
Source: I.B. Sperstad, E.H. Solvang and S.H. Jakobsen et al. , «Data set for power system reliability analysis using a four-area test network “, Data in Brief 33 (2020) 106495

Market and grid simulation - Nodal dispatch

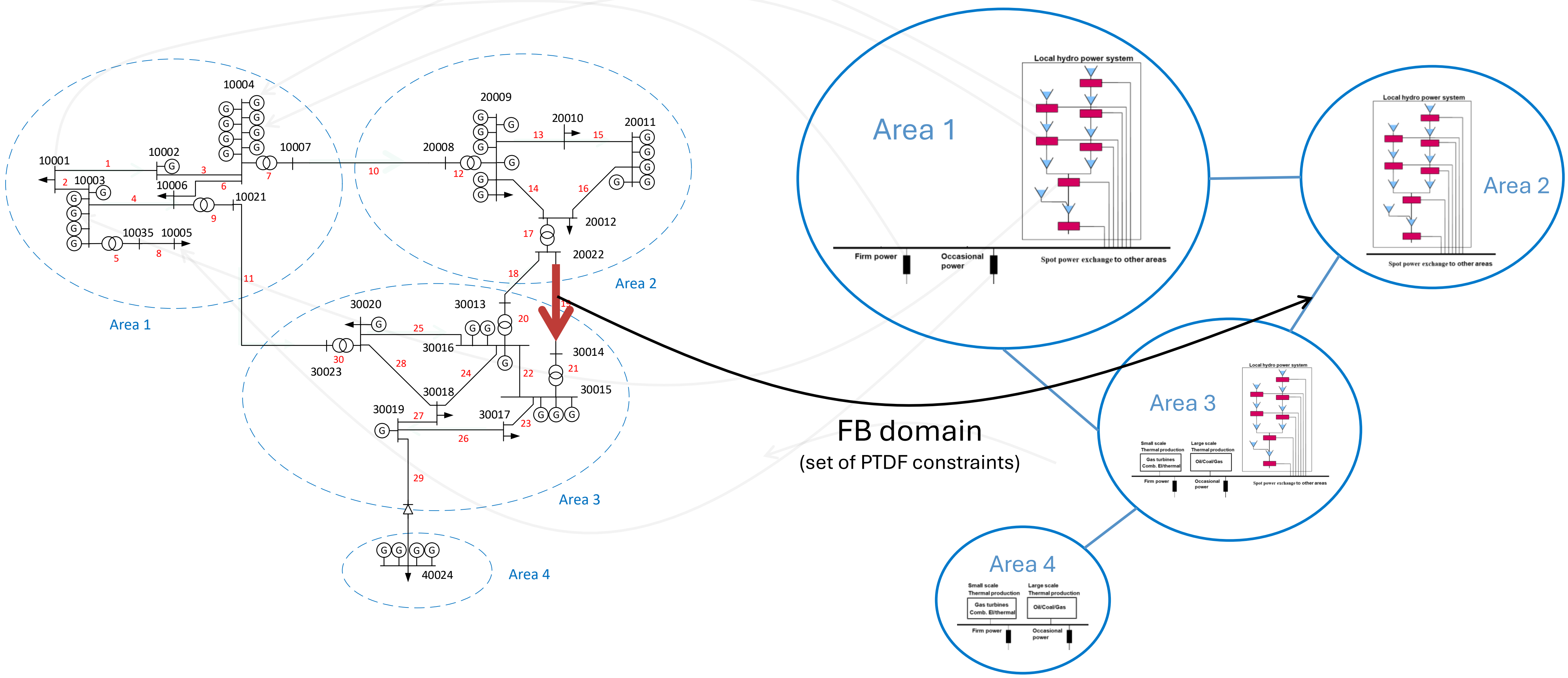


Market and grid simulation - DC power flow

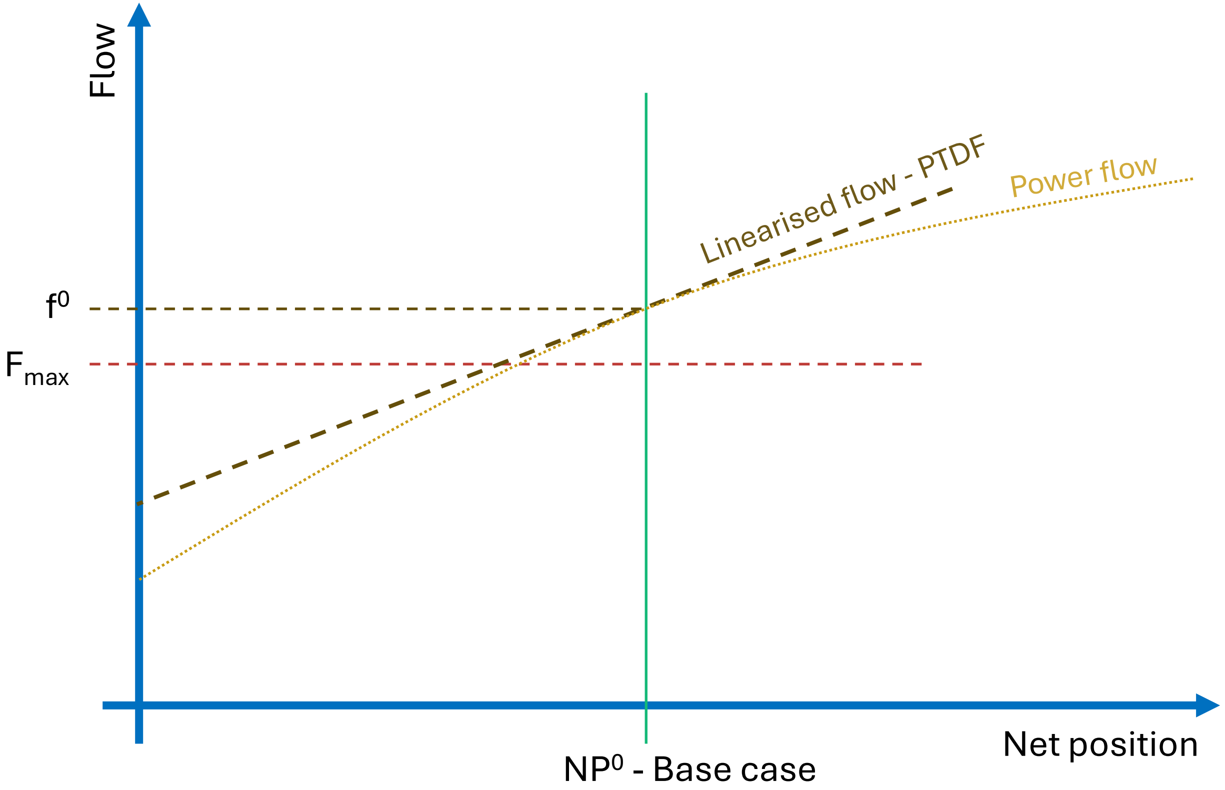




Market and grid simulation - FB domain



Calculation of flow-based constraints

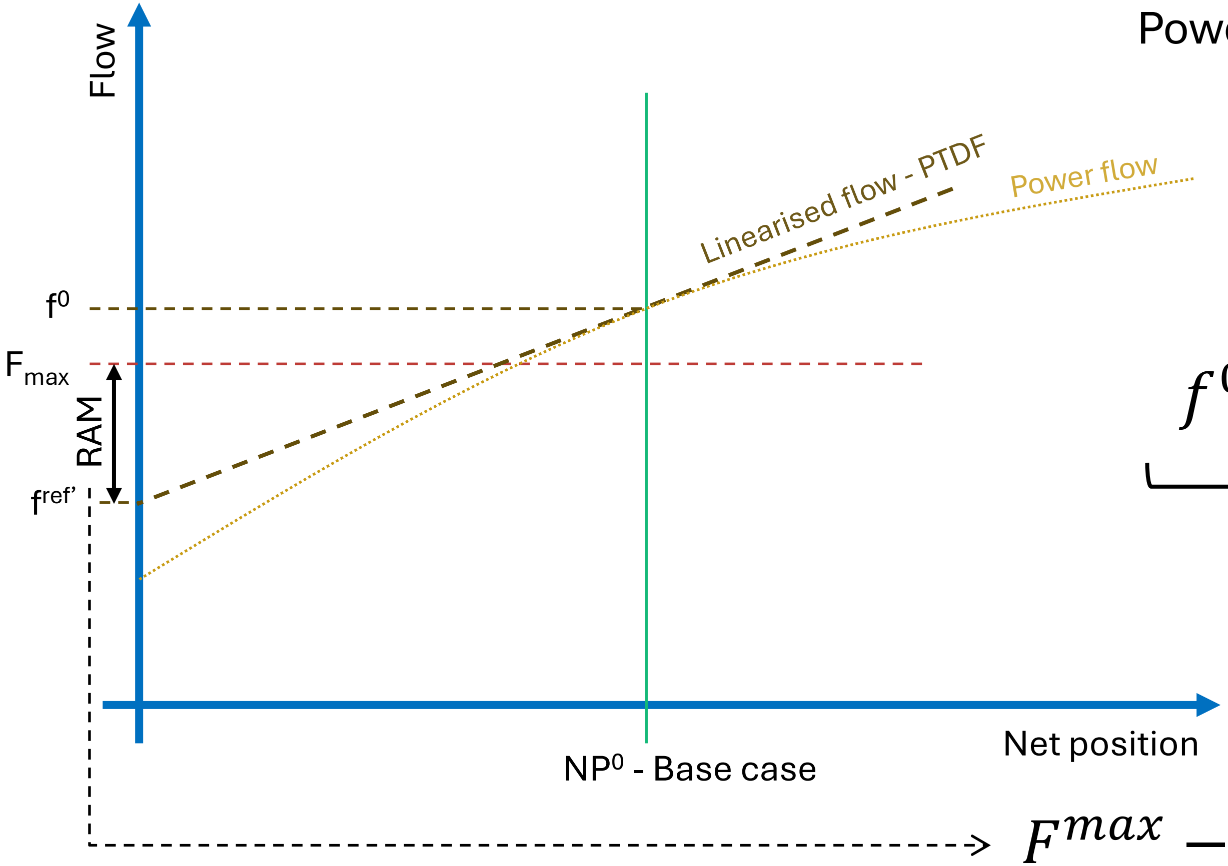


- Power Transfer Distribution Factors (PTDF) based on simplified DC representation of grid
- Distribution of system dispatch in base case to individual busses
- Power flow for base case
- Linearisation of flow in NP^0 (base case):

$$\Delta F = PTDF \times \Delta NP$$

$$F - f^0 = PTDF \times (NP - NP_0)$$

Calculation of flow-based constraints



Power-flow in base case

Net position in base case

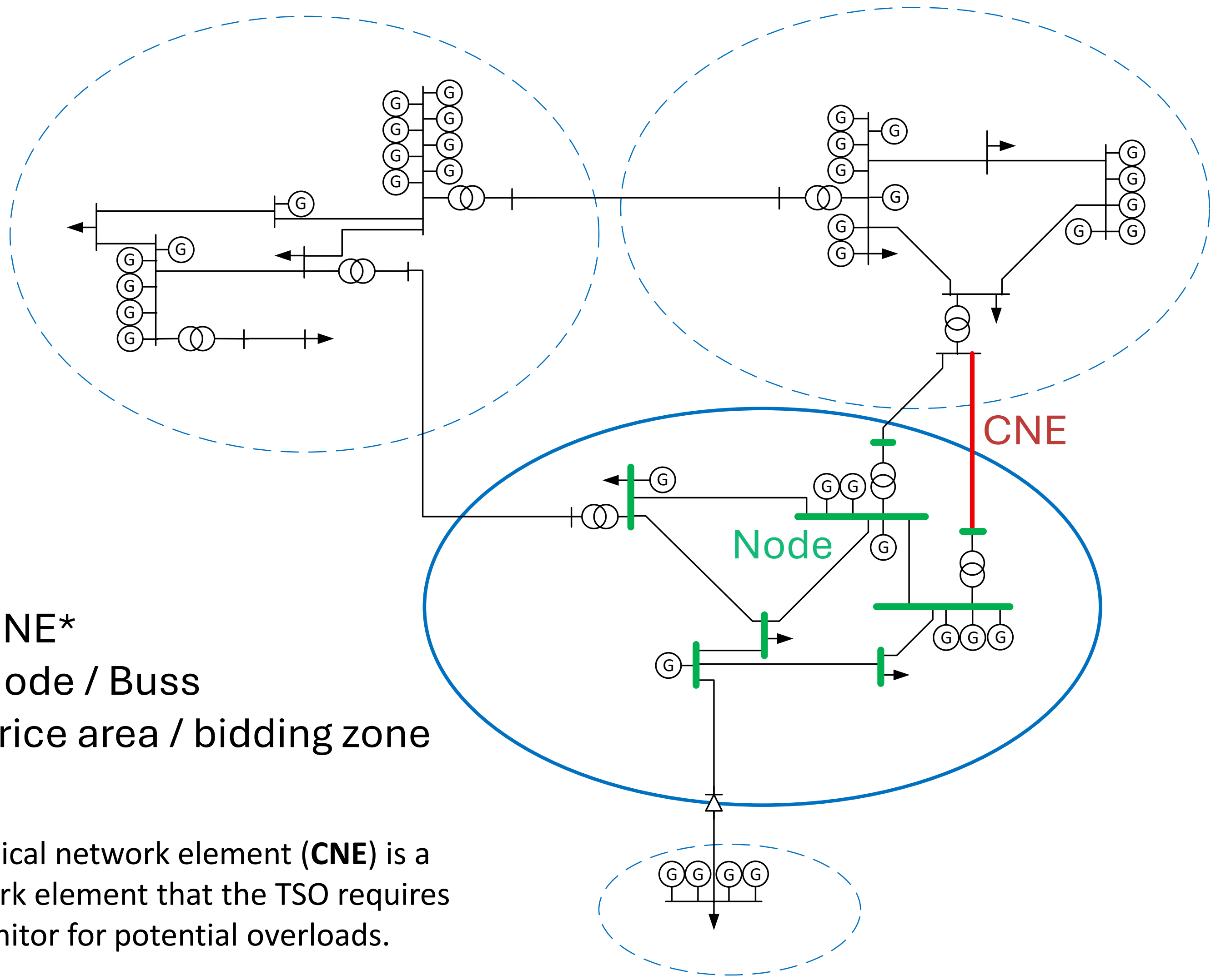
$$f^0 + PTDF \times (NP - NP^0) \leq F^{max}$$

$$\underbrace{f^0 - PTDF \times NP^0}_{} + PTDF \times NP \leq F^{max}$$

$$f^{ref\prime} + PTDF \times NP \leq F^{max}$$

$$F^{max} - f^{ref\prime} = RAM \quad \Rightarrow \quad \boxed{PTDF \times NP \leq RAM}$$

From nodal to zonal representation



c – CNE*
n – Node / Buss
a – Price area / bidding zone

*A critical network element (**CNE**) is a network element that the TSO requires to monitor for potential overloads.

✓ Derived from detailed grid:

$$PTDF_c^n \times \Delta NP_n = \Delta f_{c,n}$$

✓ Nodal PTDF

- How does the flow change on a CNE c due to a change of net position in node n

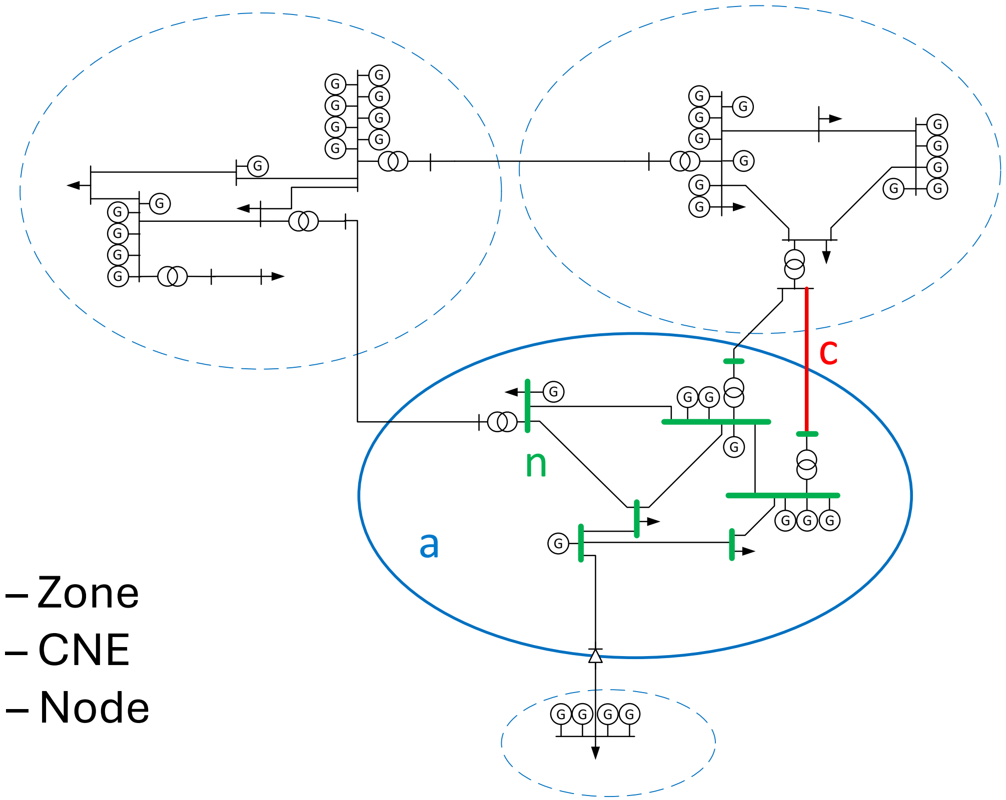
? Necessary for zonal market clearing:

$$PTDF_c^a \times \Delta NP_a = \Delta f_{c,a}$$

? Zonal PTDF

- How does the flow change on a CNE c due to a change of net position in zone a

From nodal to zonal PTDFs using Generation Shift Keys



a – Zone
c – CNE
n – Node

The Generation Shift Key (**GSK**) is a parameter used in the translation from node-to-CNE PTDFs to zone-to-CNE PTDFs.

$$\sum_{n \in N_a} GSK^n \times PTDF_c^n = PTDF_c^a$$

$PTDF_c^a$	PTDF for zone a on CNE l ("Zonal PTDF")
GSK^n	GSK for node n
$PTDF_c^n$	PTDF for node n on CNE l ("Nodal PTDF")
N_a	Nodes in zone a

Simple / flat GSK strategy: $GSK^n = \frac{1}{N}$

Net position GSK strategy: $GSK^n = \frac{|NP_n^o|}{\sum_{m \in N_a} |NP_m^0|}$

In our initial version of the FB procedure, we are opting for a flat GSK strategy. However, the outcome of the flow predictions from different GSK strategies will be monitored over time, which will give a base for developing a potential better strategy in a later version of the Nordic FB.

Source: Methodology and concepts for the Nordic flow-based market coupling approach

From nodal PTDFs to zonal FB constraints

$$\sum_{a \in A} PTDF_c^a \times NP_a \leq F_c^{max} - f_c^0 + \sum_{a \in A} PTDF_c^a \times NP_a^0$$

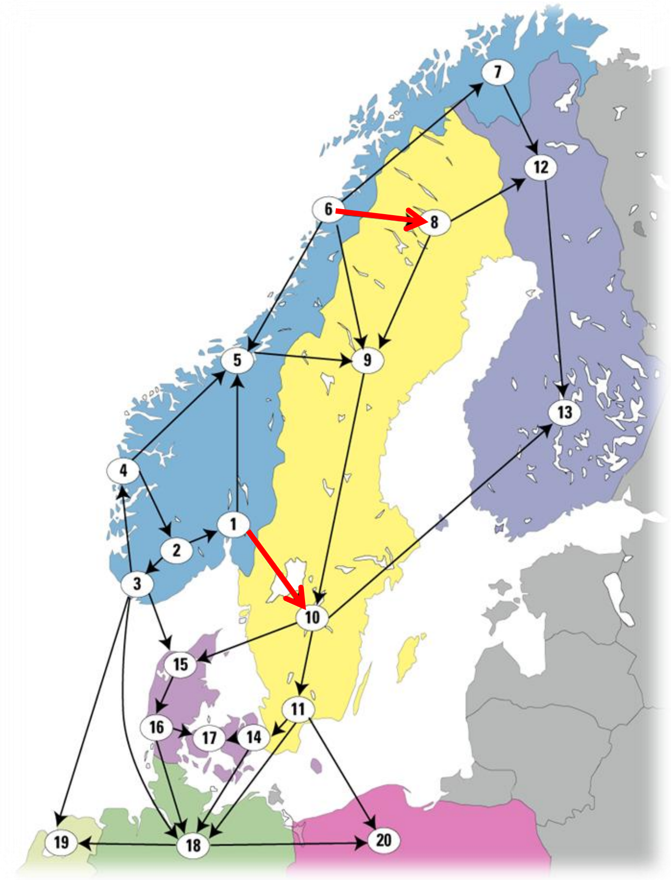
$$\sum_{a \in A} NP_a \times \sum_{n \in N_a} GSK_c^n \times PTDF_c^n \leq F_c^{max} - f_c^0 + \sum_{a \in A} NP_a^0 \times \sum_{n \in N_a} GSK_c^n \times PTDF_c^n$$

$$\overrightarrow{NP} \quad \times \quad \overrightarrow{PTDF_c} \quad \leq \quad RAM_c$$

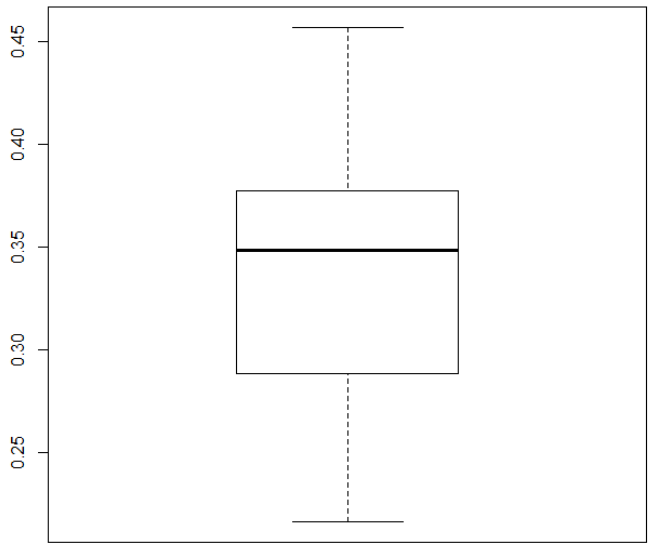
Zonal PTDF is depending on GSK strategy

RAM is depending on GSK strategy

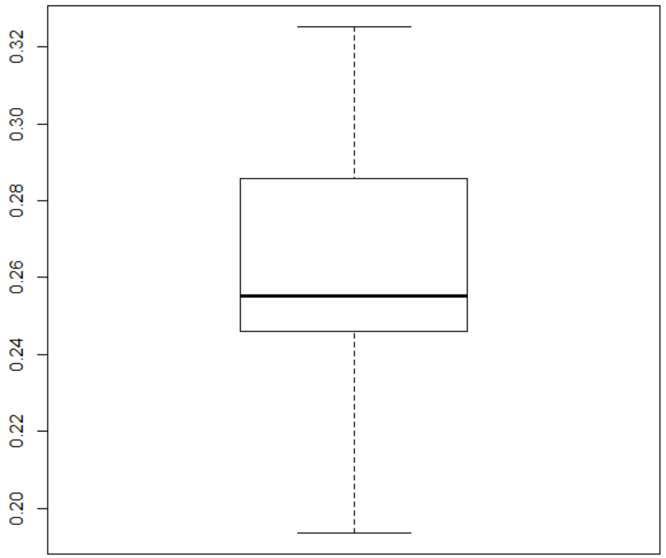
Example for distribution of nodal PTDFs within an area



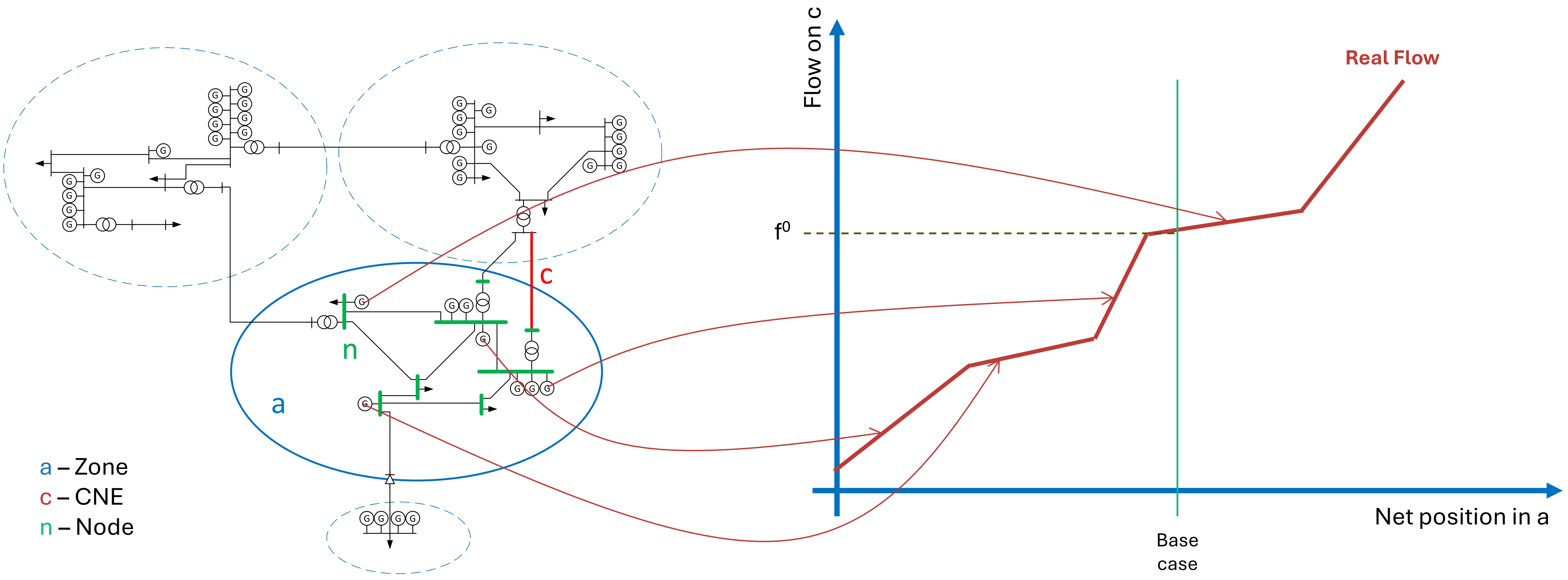
Nodal PTDFs from nodes in area
Nordland (6) on
CNE Nordland -> Nord-Sverige (6->8)
(114 nodes)



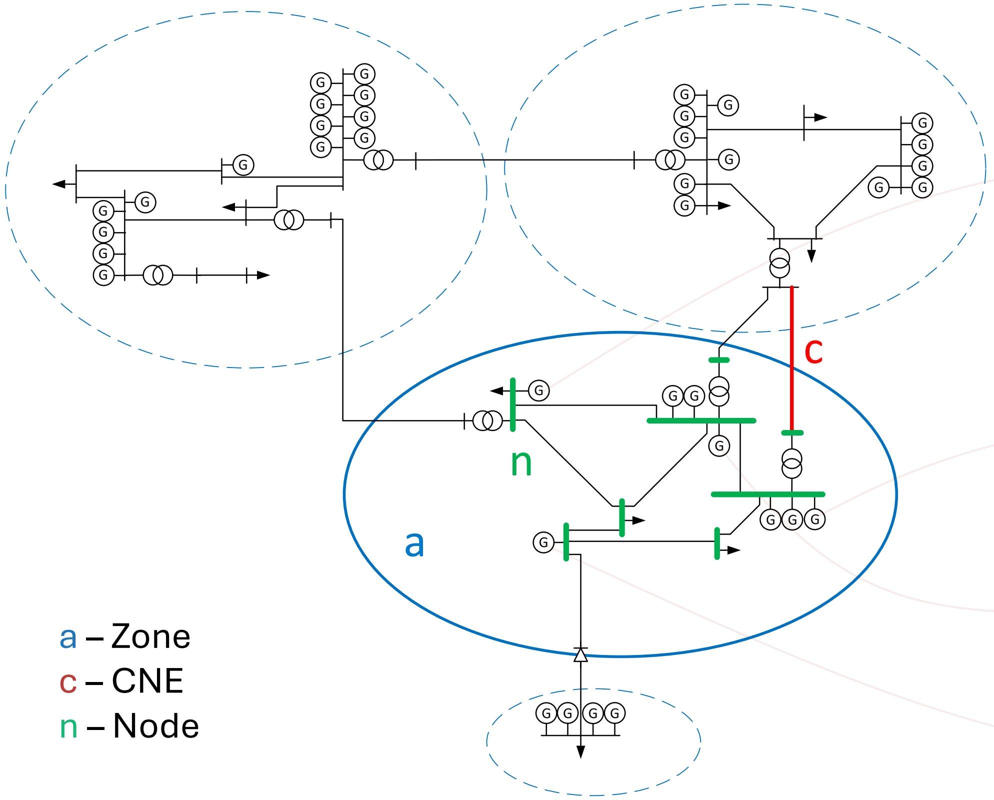
Nodal PTDFs from nodes in area
Mid-Norway (5) on
CNE Østland -> Central Sweden (1->10)
(125 nodes)

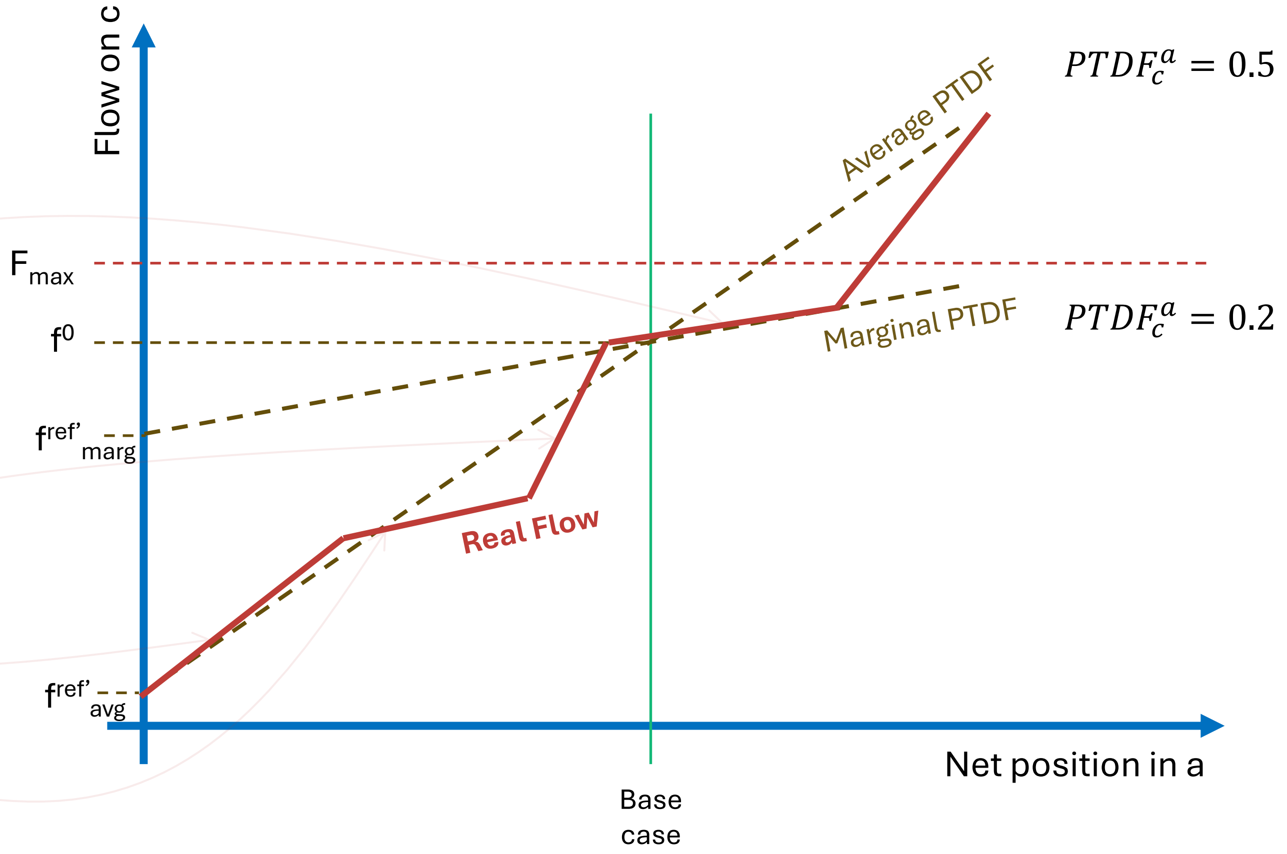


Nodal contribution to flow

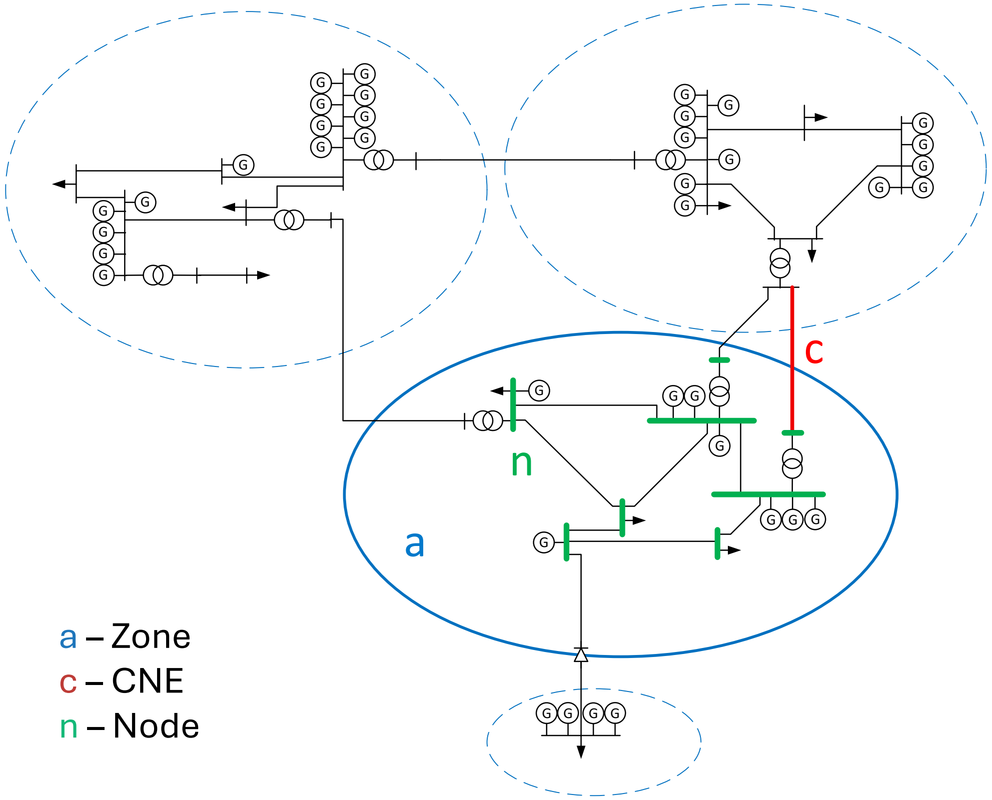


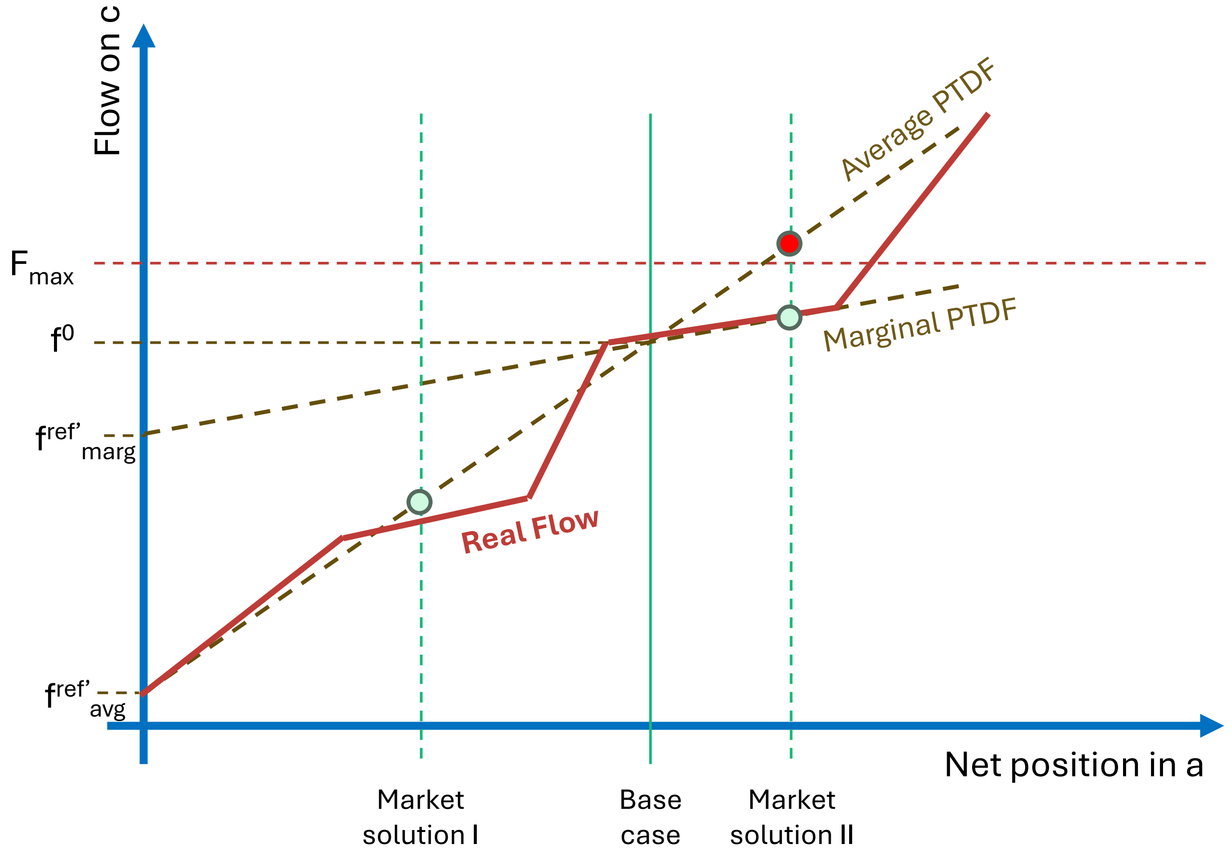
Nodal contribution to flow





Nodal contribution to flow and market solutions

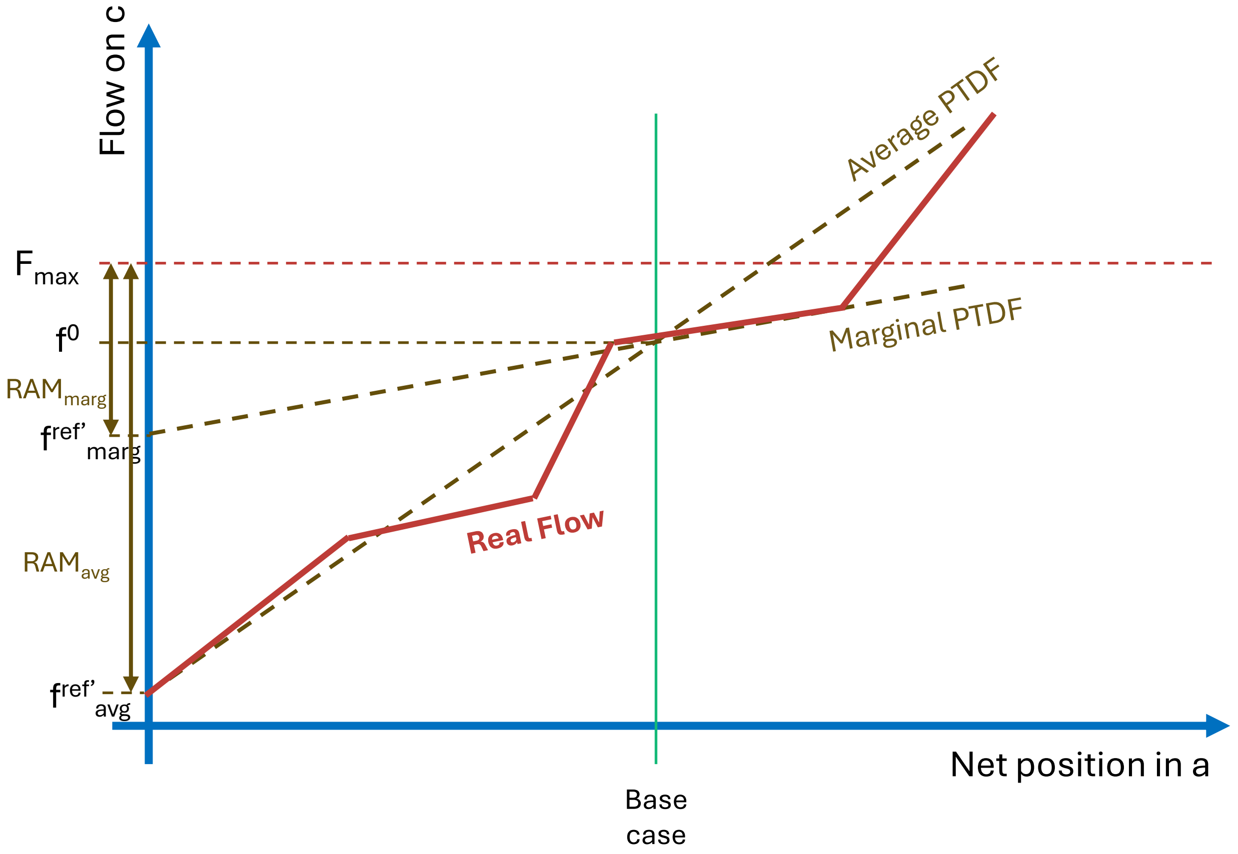




Impact of GSK strategy for nodal to zonal aggregation

- Generation Shift Keys (GSKs) estimate how each asset (production / consumption) node contribute to a change in flow
- Nodal PTDFs vary throughout each price area and
- Marginal units that react on changes in net position are geographically distributed
- Zonal PTDFs and RAM depend on GSK strategy

✓ **There is no theoretically correct method to define GSKs. The choice of GSKs influences the FB domain and market outcome.**



SINTEF

Suggested further readings


Methodology and concepts for the Nordic Flow-Based Market Coupling Approach

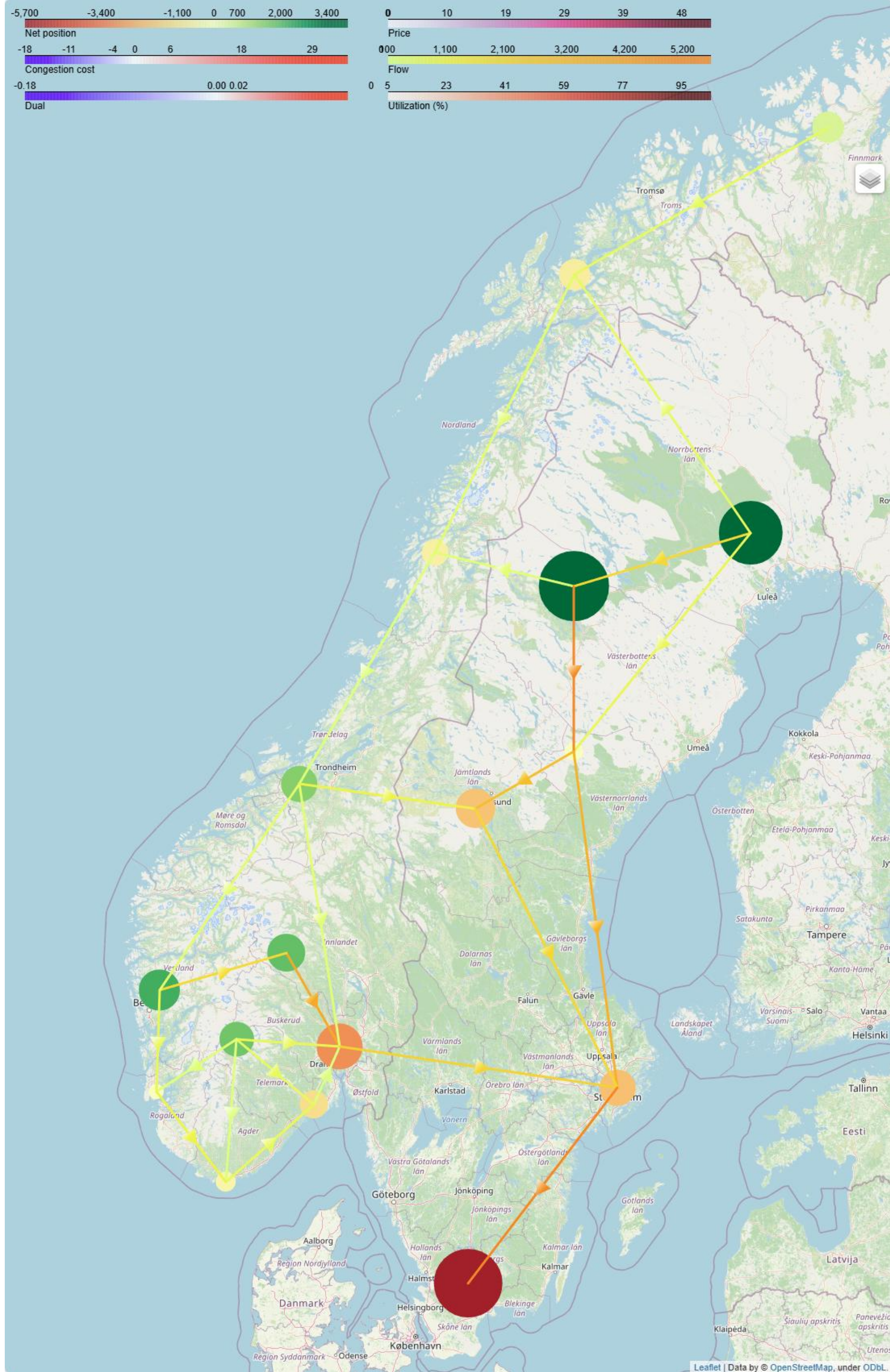
ENERGINET DK SVENSKA KRAFTNÄT FINGRID Statnett

Analyses of GSK strategies

and their consequences

- ➢ Flow based domain consists of:
- ➢ PTDF
  - ➢ Depending on grid topology => rather stable
- ➢ RAM
  - ➢ Depending on base case => changes with market clearing

✓ Forecasting of PTDFs and RAM is essential for long-term power market modelling



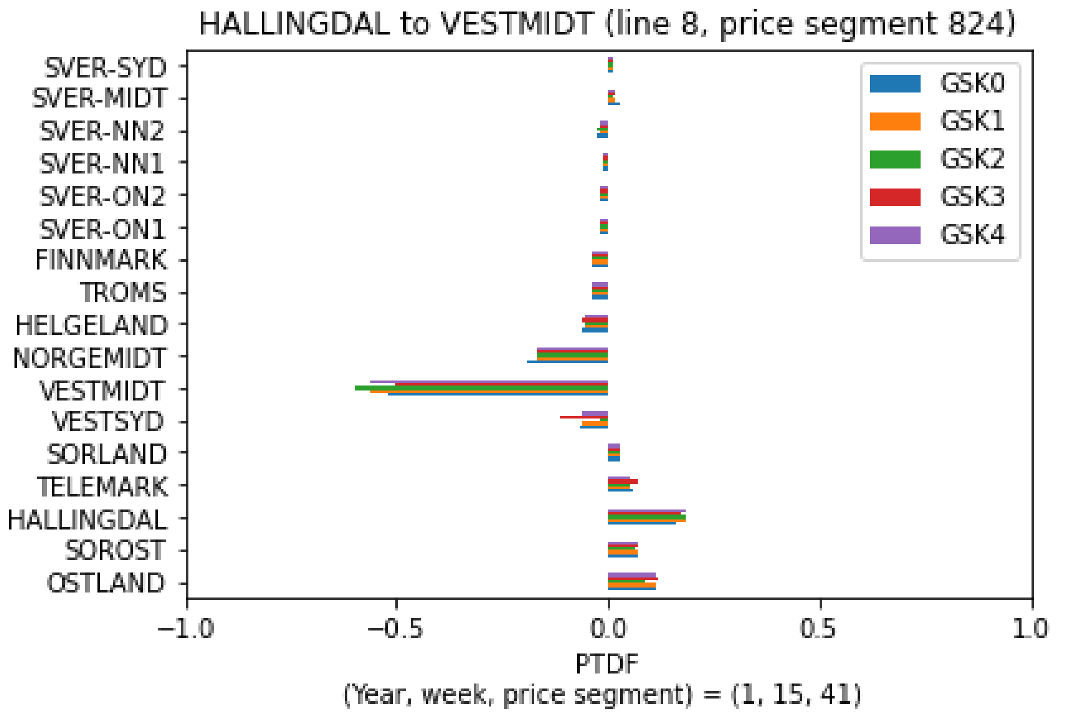
Analysed GSK strategies

GSK	strategy	comment
0	flat	Flat contribution of all nodes
1	net position	Net injection in a node (which can be zero)
2	generation	Net generation in a node
3	load	Net consumption in a node
4	generation + load	Net generation plus net consumption in a node
…		there are further and even more advanced strategies

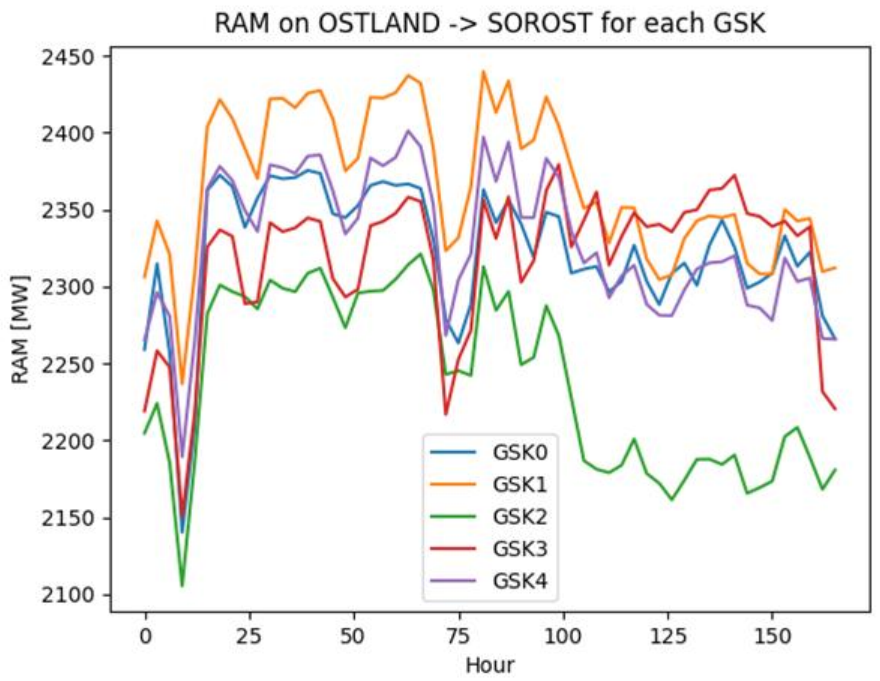
Effect of GSK strategies on PTDFs and RAM

GSK	strategy
0	flat
1	net position
2	generation
3	load
4	generation + load

PTDF



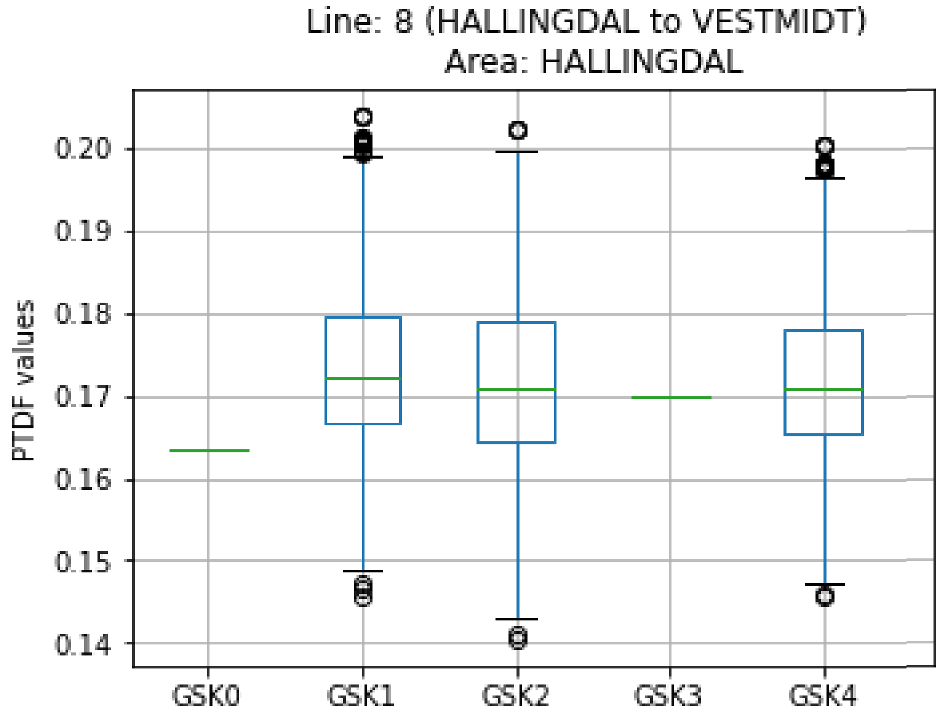
RAM

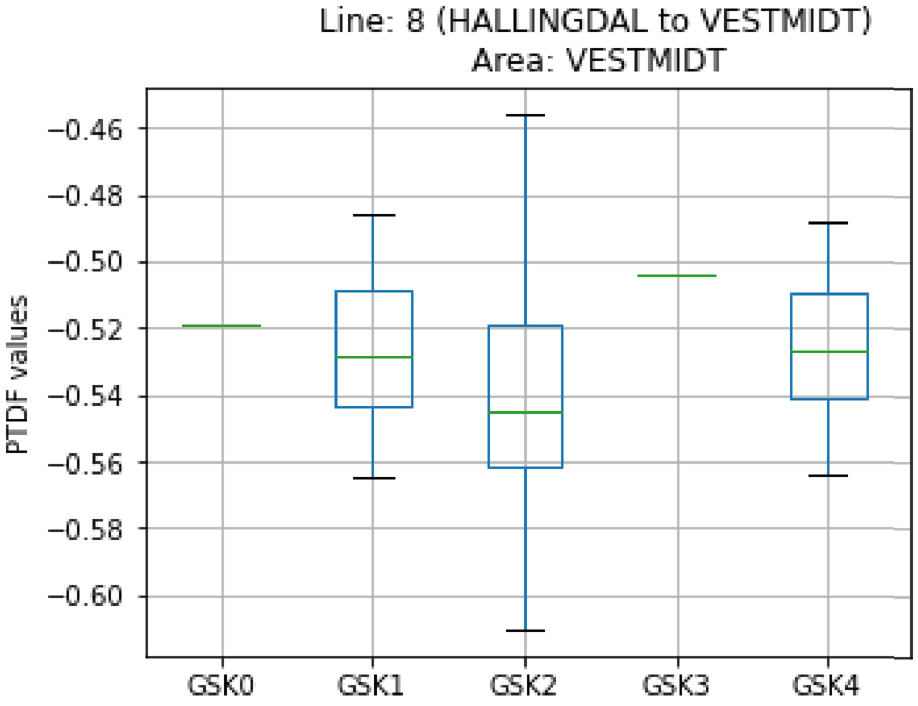


Distribution of PTDFs

in some strategies these are depending on the dispatch

GSK	strategy
0	flat
1	abs(net position)
2	abs(generation)
3	abs(load)
4	abs(gen)+ abs(load)





Short summary

- Power markets are cleared based on an area (zonal) description, while the physical power flow is distributed over a grid with numerous busses (nodes) and lines.
- The Generation Shift Key (GSK) is a value used in the translation from node-to-CNE PTDFs to zone-to-CNE PTDFs and define how each production / consumption node contribute to a net position change in an area.
- Choice of GSK affects PTDFs and RAM, but has limited effect on results, as RAM is adapted to compensate for different zonal PTDFs.
- **There is no theoretically correct method to define GSKs. The choice of GSKs influences the flow-based domain and market outcome.**

Teknologi for et bedre samfunn